

1 Motivation

Masked diffusion LLMs generate text by iteratively unmasking tokens *in parallel*, enabling any-order planning. Yet current inference relies on **static confidence thresholds** and **fixed block schedules**:

- ▶ Uniform compute regardless of position difficulty.
- ▶ No adaptive self-correction of mis-drafted tokens.
- ▶ Per-task hyperparameter tuning required.

Jazbec et al. (2025) trained a lightweight RL policy for *unmasking* via cold-start GRPO, but this covers only unmasking and collapses in our dual-model setting (10–14% accuracy after 150 RL steps; see §4).

Goal: a lightweight policy jointly controlling **unmasking** (u_t) and **remasking** (r_t) on top of a speculative dual-mode inference loop trained stably without cold-start RL instability.

2 Dual-Mode MoE Architecture

LLaDA 2.1-mini is a 16B-parameter MoE with 256 routed experts per layer. AOAExposes **two operational modes of the same frozen model**:

Hard-routed drafter (fast)

Standard top-8 routing $\Rightarrow \sim 1.4\text{B}$ active params.
Cheap draft microsteps proposing token values for all masked positions.

Soft-routed verifier (expressive)

Temperature-widened routing to top-16 experts $\Rightarrow$ **all 16B params participate**. Periodic validation; applies PRISM-style keep / [M] decisions at each verifier event. Accepts position k when drafter–verifier argmax agree $\geq \delta_{\text{acc}}$; remasks otherwise. Both modes share the *same frozen weights*: routing toggled in-place, halving memory versus loading two separate model copies.

Inference loop (per verifier event t):

- Draft:** hard-routed pass; policy decides which masked positions to unmask ($u_t^k=1$).
- Verify:** soft-routed pass; low-agreement positions get $r_t^k=1$ (remask to [M]).
- Commit:** accepted positions join the response; rejected positions re-enter the draft pool.

3 Lightweight Policy ($\sim 465\text{K}$ params)

PhaseAV2Policy: 1-layer bidirectional transformer ($d=128$, 4 heads). Per-position scalar inputs:

Feature	Dim	Description
Mask indicator m_t^k	1	Position currently masked?
Step ratio t/T	1	Diffusion progress
Max confidence c_t^k	1	$\max_v p_\theta(v \mid \mathbf{y}_t)$
Agreement α_t^k	1	Drafter $\leftrightarrow$ verifier argmax match
PRISM score q_t^k	1	Verifier quality; 0 on drafter steps
Total (V4)	5	Scalar-only
Total (V5)	5+emb	+ h_{final} for r_t head

Active heads: $u_t^k \sim \text{Ber}(\sigma(f_\phi^u(s_t)^k))$ (unmask), $r_t^k \sim \text{Ber}(\sigma(f_\phi^r(s_t)^k))$ (remask). Cache and access heads disabled; reserved for future work.

4 Two-Stage Training Pipeline

Stage 1 Supervised Warmstart (dense BCE)

Run speculative rollouts with the heuristic policy. For each position k per trajectory step, derive binary labels from future verifier outcomes:

$$\ell_t^{u,k} = \mathbf{1}[\text{verifier would accept}], \quad \ell_t^{r,k} = \mathbf{1}[\text{verifier would reject}].$$

Train with per-position binary cross-entropy **dense signal, dense gradients**. **500 steps, ~ 2.5 h on $1\times\text{H200}$** . Final BCE loss ≈ 0.002 ; smooth convergence, no instability.

Stage 2 GRPO (Reinforcement Learning)

Initialise from warmstart checkpoint. Sample $G=4$ rollouts per prompt; compute group-mean advantages; update with clipped PPO surrogate ($\varepsilon=0.2$). **No expert steering. No KL penalty.**

Reward (multiplicative prevents reward hacking):

$$R = r(\hat{\mathbf{y}}, \mathbf{y}^*) \cdot (1 - \text{eff_flops})^\alpha - \beta \frac{\text{remask}}{L} - w_u f_{\text{unresolved}}$$

where $\text{eff_flops} = \text{used_steps} / T$, $\alpha=1$.

200 steps (V4); $4\times\text{H200}$, DP=4.

5 Key Insight: Why Warmstart Enables GRPO

	Cold-start RL (V3)	Warmstart $\rightarrow$ RL (V4/V5)
Initialisation	Random	Near-good solution
Training signal	Sparse reward only	Dense BCE $\rightarrow$ sparse reward
Expert steering needed?	Yes (still fails)	No
Training stability	Oscillates; stalls	Smooth convergence
GSM8K accuracy	10–14%	86%

Jazbec et al. tried conditioning the policy on hidden state h_t from cold-start RL and found it *hurt*: sparse end-of-episode reward cannot navigate high-dimensional hidden-state space. **PRISM** (Kim et al., 2025) showed the same hidden states *are* learnable via dense supervised labels. Our warmstart is structurally analogous to PRISM fine-tuning: dense per-position BCE labels solve credit assignment first; GRPO then refines for the global correctness $\times$ speed objective.

Analogy: cold-start RL on h_t is like training a medical classifier from scratch on 100 labels; warmstart is like fine-tuning an ImageNet-pretrained backbone the prior structure does the heavy lifting.

6 V5: Head-Specific Feature Conditioning

Head	Microstep	Extra input	Rationale
u_t (unmask)	Drafter only	$\mathbf{e}_t$ (soft-masked emb.)	Token-identity signal; free
r_t (remask)	Verifier only	h_{final} (last hidden)	PRISM-style; verifier already running
Gates initialised to -8 ($\approx 0.03\%$ contribution) so warmstart starts identically to scalar-only; they open gradually under supervised gradient pressure. V5 GRPO evals in progress (warmstart: 78.2% GSM8K at 44.8 TPS, 1319 samples).			

7 GSM8K Results

Development evals (50 samples; lossless verification, agree = 0.5, 256 denoising steps):

Method	Training	Steps	τ	Acc.	TPS
<i>No-training baselines</i>					
DefaultPolicy	heuristic		1.0	84%	53.1
<i>V3 Jazbec-style cold-start GRPO + expert steering</i>					
Cold-start GRPO	RL only	150	0.5	14%	49.4
Cold-start GRPO	RL only	150	1.0	10%	18.8
<i>V4 Supervised warmstart (Stage 1 only)</i>					
Warmstart	BCE	500	0.5	72%	44.1
Warmstart	BCE	500	1.0	84%	31.1

V4 Warmstart $\rightarrow$ GRPO (full pipeline)

W/S + GRPO	BCE $\rightarrow$ RL	500+100	0.5	70%	60.7
W/S + GRPO	BCE $\rightarrow$ RL	500+100	1.0	66%	57.6
W/S + GRPO (best)	BCE $\rightarrow$ RL	500+200	0.5	86%	57.3
W/S + GRPO	BCE $\rightarrow$ RL	500+200	1.0	62%	58.6

Full-scale evals (GSM8K 1,319 samples; $1\times\text{H200}$, scoring: gsm8k.llada.flexible):

Method	Training	τ	Lossless	Acc.	TPS
LLaDA speed_mode				74.6%	72.6
LLaDA quality_mode				77.3%	64.7
DefaultPolicy		1.0	off	80.7%	46.0
V5 warmstart hybrid	BCE	0.5	off	78.2%	44.8
V4 W/S + QBal GRPO (best)	BCE $\rightarrow$ RL	0.5	off	79.6%	72.6
V4 W/S + QBal GRPO	BCE $\rightarrow$ RL	0.5	on	76.0%	77.2

W/S = warmstart. QBal = quality-balanced operating point (agree=0.92), 625 GRPO steps. V5 GRPO evals pending.

8 Key Findings

- Cold-start GRPO (V3) fails.** Expert steering (Jazbec et al.’s stabiliser) still yields only 10–14% GSM8K accuracy after 150 RL steps. Training correctness never rises above 0.17.
- Warmstart alone recovers baseline accuracy.** 500 BCE steps reach 84% accuracy, matching the training-free DefaultPolicy. The policy has learned meaningful per-position decision boundaries.
- Warmstart + GRPO Pareto-improves over all baselines.** 50-sample: **86% / 57.3 TPS** vs. 84% / 53.1 (DefaultPolicy). Full-scale: **79.6% / 72.6 TPS** — quality-mode accuracy at speed-mode throughput.
- GRPO teaches compute-awareness.** Accuracy at $\tau=0.5$ jumps 70% $\rightarrow$ 86% from step 100 to 200 while TPS holds at 57–61. The multiplicative reward teaches *selective* unmasking rather than uniformly fast unmasking.